

UNIVERSITY OF EDINBURGH
COLLEGE OF SCIENCE AND ENGINEERING
SCHOOL OF INFORMATICS

INFR10074 COMPUTER COMMUNICATIONS AND NETWORKS

Monday 29th April 2019

14:30 to 16:30

INSTRUCTIONS TO CANDIDATES

Answer any TWO of the three questions. If more than two questions are answered, only QUESTION 1 and QUESTION 2 will be marked.

All questions carry equal weight.

CALCULATORS MAY BE USED IN THIS EXAMINATION

Year 3 Courses

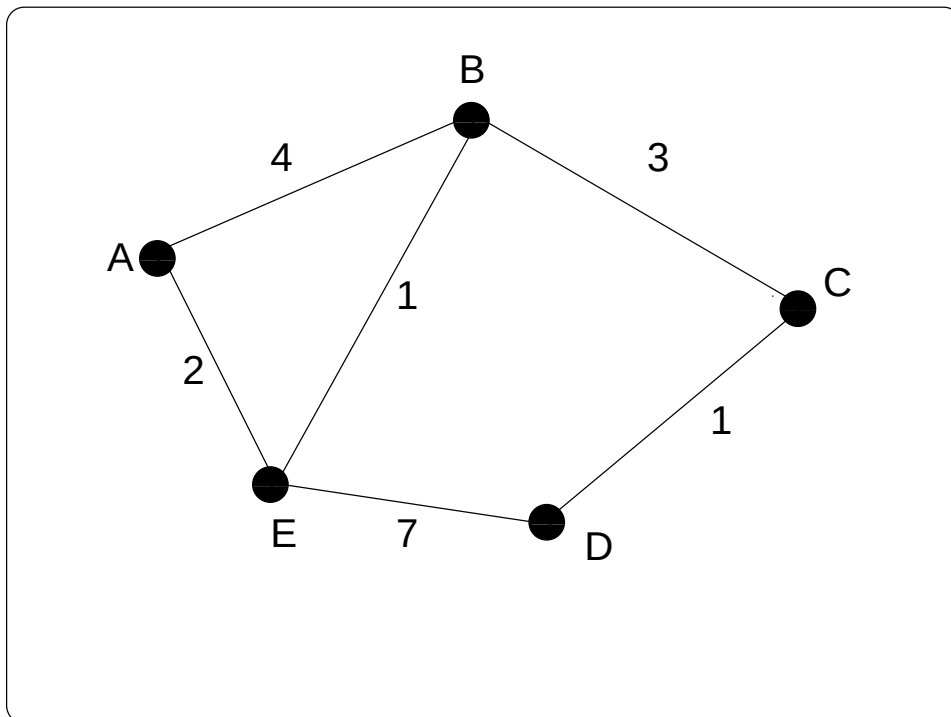
Convener: C. Stirling

External Examiners: S.Rogers, S. Kalvala, H.Vandierendonck

THIS EXAMINATION WILL BE MARKED ANONYMOUSLY

1. (a) Compare cable and ADSL based broadband access networks. [3 marks]
- (b) Suppose you acquired a new domain for your organisation called mynewdomain.org. How would the DNS system be made aware of it so that any requests to access www.mynewdomain.org are received by the web server of your domain? [3 marks]
- (c) Give three reasons why HTTP streaming is preferred over UDP streaming for video streaming applications. [3 marks]
- (d) With HTTP streaming, are the TCP receive buffer and the media player client's application buffer the same thing? If so, why? If not, how do they interact? [2 marks]
- (e) The high-quality version of an hour long TV programme on BBC iPlayer has a size of 350MB to download to a home computer. A home user has a broadband service with a download speed of 2Mbps. Round-trip time to the BBC iPlayer website is found to be 100ms using a ping test with default packet size of 56bytes. Assume that the access link is the bottleneck and delays other than those due to propagation and transmission are negligible. Calculate the time it takes for the user to download the whole programme and the throughput of the transfer. How much does the propagation delay matter for this transfer? [6 marks]
- (f) Consider distributing a file of $F = 10$ Gigabits (Gb) to N peers. The server has an upload rate of $u_s = 50$ Mbps, and each peer has a download rate of $d_i = 2$ Mbps and an upload rate of u . For $N = 10$ and 100 , and $u = 250$ Kbps and 500 Kbps, tabulate the minimum distribution times for each of the combinations of N and u for both client-server distribution and P2P distribution. [8 marks]

2. (a) Distinguish between control and data plane at the network layer. Mention two different paradigms to realise the network layer control plane and the same for the data plane. [4 marks]
- (b) Describe the OpenFlow data plane abstraction. [2 marks]
- (c) List five key differences between IPv4 and IPv6. [5 marks]
- (d) Suppose a router has the following CIDR entries in its routing table:
- | Net/Prefix | Next Hop |
|----------------|-------------|
| 135.46.56.0/22 | Interface 0 |
| 135.46.60.0/22 | Interface 1 |
| 192.53.40.0/23 | Router 1 |
| default | Router 2 |
- For each of the following IP addresses, what does the router do if a packet with that address arrives? (i) 135.46.63.10; (ii) 135.46.57.14; (iii) 135.46.52.2; (iv) 192.53.40.7; (v) 192.53.56.7. [5 marks]
- (e) Describe a general mechanism to resolve the counting-to-infinity problem. [5 marks]
- (f) The figure shown below indicates the state of the network topology and link costs at node *A* at the end of flooding-based link state dissemination (i.e., using link-state broadcast algorithm). Determine the shortest path from node *A* to every other node using Dijkstra's shortest path algorithm. Show all steps of the algorithm.



[4 marks]

3. (a) Describe two scenarios where use of standard TCP can result in unfairness. [2 marks]
- (b) One of the desired properties of multiple access protocols is fairness. Compare and contrast different approaches to multiple access in terms of the fairness property. [3 marks]
- (c) As each network interface at a router or host is assigned a unique IP address, why is a separate MAC address (also called physical address) needed? [2 marks]
- (d) How are frame losses handled by Ethernet? [3 marks]
- (e) Discuss the significance of CSMA/CD multiple access protocol in the context of switched Ethernet. [1 mark]
- (f) Consider the effect of using slow start on a link with a 20ms round-trip time and no congestion. The receive window is 16KB and the maximum segment size is 1460 bytes. How long does it take before the first full window can be sent in one transmission round? [2 marks]
- (g) Consider sending a large file from one host to another over a TCP connection that has no loss. Suppose TCP uses AIMD for its congestion control *without* slow start. Assume *CongWin* increases by 1 MSS (maximum segment size) every time a batch of ACKs is received. Also assume approximately constant RTTs (round-trip times).
- How long does it take for *CongWin* to increase from 1 MSS to 5 MSS (assuming no loss events)? [2 marks]
 - What is the average throughput (in terms of MSS and RTT) for this connection over that period (from sub-part (i))? [2 marks]
- (h) When a file is transferred between two computers, two acknowledgement strategies are possible. In the first one, the file is chopped up into packets, which are individually acknowledged by the receiver, but the file transfer as a whole is not acknowledged. In the second one, the packets are not acknowledged individually, but the entire file is acknowledged when it arrives. Discuss the relative merits of these two approaches. [4 marks]
- (i) Suppose the data portion (including the headers) of a packet contains a sequence of 8 bytes, each corresponding to the 8-bit unsigned binary representation of the following integers: 243, 81, 128, 64, 32, 16, 27, 9. The corresponding sequence of hex values for those integers is: 0xf3, 0x51, 0x80, 0x40, 0x20, 0x10, 0x1b, 0x09. Compute the Internet checksum for this data. [4 marks]